

ASTRO-DODGE

Dodge the asteroids and survive!

Table of Contents

Table of Contents.....	1
1.0 Game Overview.....	2
1.1 System Requirements.....	2
1.2 Installing the Environment.....	3
1.3 Setting up the Game.....	4
1.4 Game Controls and Powerups.....	5
1.5 Playing the Game.....	7
1.6 Game Customizations.....	7
1.7 Game Over and Restart.....	8
1.8 Troubleshooting and FAQ.....	8
1.9 Bug Report.....	8
2.0 Credits and Acknowledgments.....	9

1.0 Game Overview

Astro-Dodge will push you to your limits, testing both your ability to react quickly and how well you can think strategically when it comes to survival. Piloting an advanced, extremely nimble spacecraft as an astronaut, your goal is simply to last as long as you can while flying through an asteroid field that stretches infinitely out into the vastness of outer space.

There is no finish line or destination in this game; rather, your sole purpose is to stay alive for as long as you can and accumulate as many score points as possible. Each second that you avoid being destroyed will directly affect your score. Your best scores will automatically be recorded for competitive purposes, so you will always have something to strive for as you work on perfecting both your offense and defense capabilities to try to beat your best score.

1.1 System Requirements

Operating System: Windows 7/8/10, macOS 10.13+, or a Linux distribution (Ubuntu 20.04+).

Processor: Intel Core i3 or equivalent (Dual-core 2.0GHz).

Memory (RAM): 2 GB RAM.

Graphics: Integrated graphics with support for OpenGL 2.1 or higher.

Storage: At least 20 MB available space.

Development Environment: 1.2 GB of available space.

Display: 1920x1080 resolution.

1.2 Installing the Environment

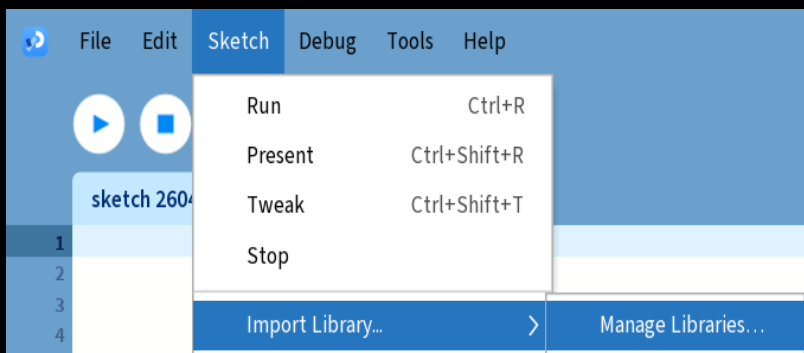
Download the Processing IDE from the internet. Install it on your device.

- [Processing Download](#)

Next, download the 'main' folder consisting of all the subfiles including: main.pde, spaceship.pde, stars.pde, asteroids.pde, gameover.pde, collision_detection.pde, Database.pde, laser.pde, shield.pde, slowTime.pde, settings.pde and the data folder.

Open the Processing IDE app and navigate to the 'Sketch' option on the top of the window.

Click on 'Import Library' and then on the 'Manage Libraries' option. The 'Contribution Manager' window should open.

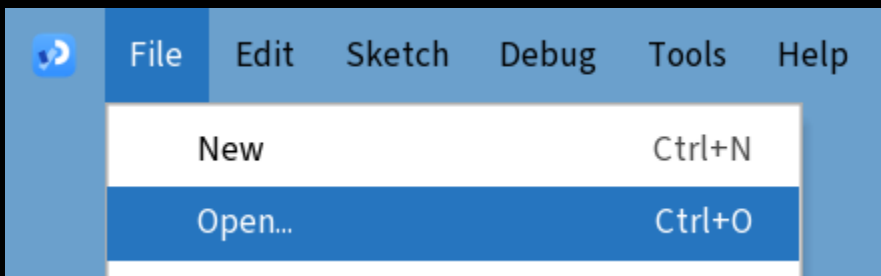


On the search bar, type 'Sound' and install the one made by 'The Processing Foundation'.

sound	All
Name ↓	Author
Cassette Implementation of Processing Sound APIs for Androi...	Shlomi Hod
ComposingForEveryone ComposingForEveryone gives support ...	Guido Kramann
Minim An audio library that provides easy to use classes for pla...	Damien Di Fede and Anderson Mills
oscP5 An Open Sound Control implementation.	Andreas Schlegel
Sound Provides a simple way to work with audio.	The Processing Foundation

1.3 Setting up the Game

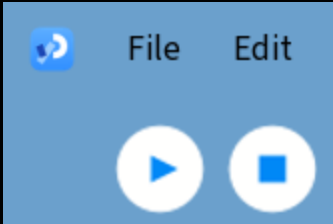
Close that window and press Ctrl+O on Windows / Cmd+O on Mac or click on 'File' on the top left corner and press 'Open'.



Navigate to the 'main' folder and click on the 'main.pde' file.

gameOver	4/20/2026 9:56 PM	Processi
home	4/20/2026 9:56 PM	Processi
laser	4/20/2026 9:56 PM	Processi
main	4/22/2026 12:41 PM	Processi
settings	4/20/2026 9:56 PM	Processi
shield	4/20/2026 9:56 PM	Processi

After the file opens. Press CTRL+R / CMD+R or click on the run icon.



The main screen of the game will be visible.



The 'Start Mission' button can be used to start the game.

The 'Settings' button can be used to customize the game to your preferences.

The 'Exit Sector' button can be used to close the game window.

1.4 Game Controls and Powerups

Movement: Move your mouse left and right to steer your spaceship and dodge the asteroids. The vertical position of the spaceship remains fixed. Dynamic tilt allows your spaceship to lean more, the faster you move your mouse.

Laser (Offense): Press and hold the 'A' key to fire a vertical beam that destroys the asteroid. It can be continuously used for 0.8 seconds while taking 3 seconds to fully replenish its charge.



Shield (Defense): Press and hold the 'S' key to equip a round shield around your spaceship that destroys the asteroid upon contact. It can be continuously used for 1 second and takes 3 seconds to fully replenish its charge.



Slow Time: Press and hold the 'D' key to slow down time and drastically decrease the asteroid's speed making it easier to maneuver through tight gaps. It can be continuously used for 0.8 seconds while taking 3 seconds to fully replenish its charge.



Pause: Press the 'P' key to pause and unpauses the game.

Mute: Press the 'M' key to mute and unmute the game music.

1.5 Playing the Game

Press the Play button to start the game.

Move the mouse along the X-Axis to navigate the spaceship.

You earn 1 point for every frame of survival (for 60hz monitors only).

Destroy an asteroid using your shield or laser to get a 50 points bonus.

As you progress, the speed of the asteroids will increase, requiring faster reflexes.

1.6 Game Customizations

Click on the 'Settings' button.

The user can select their ship's skin, the background and the accent colors from the given options.

SHIP SKIN		BACKGROUND	
SKIN 1	SKIN 2	BG 1	BG 2
MUSIC		ACCENT COLOR	
ON	OFF	BLUE	RED

Toggle for mute and unmute of the background music is also provided.

1.7 Game Over and Restart

When the space ship collides with an asteroid without an active shield, the game ends.

Your current game score will be shown in the game over screen along with the highscore.

Press the 'Space Bar' to restart the game. Your score, the asteroid's speed and the power ups will be reset.

1.8 Troubleshooting and FAQ

The game crashes on startup with an audio error.

- Download the 'Sound' Library from '1.2'.

The game is slow and lagging.

- Close all the background applications and download the latest graphics drivers.

The high score is not being saved.

- Completely extract the 'AstroDodge' Folder and ensure the 'data' sub-folder is present.

1.9 Bug Report

Database File Error

- Deletion of the save.json file will result in the loss of high score values.

Sound Overlap Issue

- The destruction of multiple obstacles simultaneously will result in overlapping of audio resulting in increased loudness.

Hitbox Alignment Problem

- Due to the tilting mechanics, some collisions may not be detected or falsely detected by sudden movements.

Refresh Rate Issue

- **Playing using higher refresh rate monitors may experience faster moving asteroids and rapid scoring compared to the standard 60hz monitors.**

2.0 Credits and Acknowledgments.

Lead Developer and Game Designer

- Sulav Bhatta

User Interface and User Experience (UI/UX)

- Sulav Bhatta

Software and Environment

- The Processing Foundation
- Java Runtime Environment

Libraries and Assets

- Processing Sound Library
- [Pixabay.com](https://pixabay.com) (sound effects)
- Canva Creative Suite
- Horizon Font

Academic Support

- Faculty of Data Science at British International College.